



SWARM DRONE CHALLENGE

Regulations

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Ingolstadt



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Preface

Welcome to the Swarm Drone Challenge!

The Swarm Drone Challenge 2026 (SDC26) is an international competition dedicated to advancing the capabilities of autonomous aerial systems through the development and demonstration of swarm intelligence. It invites students, professionals, and startups to design and implement self-developed algorithms that enable groups of drones to cooperate effectively in complex, realistic mission environments.

Building on the success of previous editions, SDC26 continues the established Capture the Flag scenario, in which two teams deploy multiple drones operating collaboratively as a swarm. This competitive format promotes diverse technical and strategic approaches, addressing key challenges such as resilient communication, AI-based coordination, adaptive decision-making, and collective mission behavior under dynamic conditions.

The purpose of the Swarm Drone Challenge is to explore novel methods for distributed autonomy, evaluate innovative algorithmic concepts, and contribute to the evolution of common frameworks and standards for cooperative unmanned systems. By fostering experimentation and knowledge exchange among participants, SDC26 aims to identify promising technological directions and advance the understanding of swarm robotics in both theoretical and applied contexts.

Changelog

Changes in version: v5.0.0

- Defined tournament structure: 6 teams compete across three stages (Opening Round, Semi-Finals, Final); the 4 best-ranked teams advance from the Opening Round and the 2 match winners advance to the Final. Pairings are set by the organizers and announced via Discord.
- Defined overall challenge scoring: maximum 25 challenge points across three categories — up to 15 from match wins (5 per won match), up to 5 from jury evaluation (pitch and overall approach), and up to 5 from the technical report.
- Defined technical report requirements: must cover localisation approach, flight control and trajectory planning strategies, and swarm strategy. Report is due 4 days before the Final. Open-sourcing code is considered positively.
- Removed the rule that target object positions vary per match.
- Confirmed challenge location: ILA Berlin, Thursday 11 June 2026.
- Differentiated between game points (scored during a match via box captures: 1, 5, or 10) and challenge points (overall tournament leaderboard: match wins, jury evaluation, technical report).
- Fixed ArUco marker positions and IDs in the arena.
- Corrected field length from 25 m to 20 m.
- Removed zero ArUco marker from positioning system.
- Restructured document into four sections: Competition Rules, Team Regulations, Safety Regulations, and Technical Regulations.
- Promoted gameplay objective to introductory prose in the Target Objects subsection.
- Split Intellectual Property and Reporting Requirements into separate subsections.
- Consolidated all safety-related rules into a dedicated Safety Regulations section.
- Merged Drone Size/Weight Limits and Drone Identification into a single Drone Requirements subsection.
- Added Game Logic section with precise capture mechanics, ArUco marker encoding table, point scoring rules (1/5/10 points), and Special Maneuver definition.

1 Competition Rules

1.1 Playing Field

The playing field is a rectangular area with the following dimensions (Figure 1):

- Length: 20 meters
- Width: 10 meters
- Height: 6 meters

The playing field is divided into three zones:

- Two Team Zones (red and blue): The home base of each team, where the drones start, land and targets are strategically located.
- One Neutral Zone in the middle: A neutral territory for strategic maneuvers.

The sizes shown in the graphic are approximate values. The exact dimensions may vary slightly.

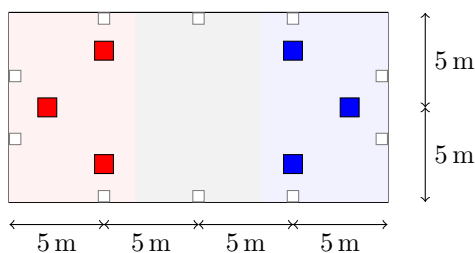


Figure 1: Field Dimensions

1.2 Target Objects

The objective of the game is to change the color of the target objects to your team's color (red or blue). All gameplay mechanics, scoring rules, and special conditions are described in detail in Section 1.4.

- Each team zone has three target objects, for a total of 6 in the playing field.
- These objects change their color based on their interaction with the drones:
 - Default state: Matches the color of the team zone.
 - Triggered state: When an enemy drone approaches and targets the box, it changes to the attacking team's color, updates the ArUco marker, and plays a horn signal. A subsequent enemy drone from the opposing team can then trigger it again.
- Dimensions of each target box:
 - Height: 55 cm (open: 73 cm), Width: 57.5 cm, Depth: 37.5 cm
- Detection: The target boxes detect approaching drones via an internal RFID tag attached to the drone. The system can reliably identify drones within a height range of approximately 1–2 meters.

1.3 Gameplay

- Teams: 2–5 drones per team (independent team sizes), ready to battle!
- Start Setup: Drones may be placed inside their designated team zone and powered on within the cage.
- Cage Clearance: Once all human operators and staff have exited the cage and the area is confirmed clear, the cage is closed.
- Final Start Signal: After the cage is secured, the final start signal is given and all drones are permitted to take off and begin flight operations.



Figure 2: Example of an SDC target box.

- Action: Drones fly around the arena, trigger target objects to change color, and score game points for their team. Full scoring rules are defined in Section 1.4.
- Defense Strategy: Teams may defend target boxes by positioning drones nearby. Strictly prohibited: physical obstruction, continuous hovering over targets, capture system interference, and all forms of **jamming or electronic warfare**. **"Sitting dead duck" behavior** — that is, passive sitting or hovering directly above the boxes — is not allowed and will lead to **disqualification**.
- Time Limit: The game lasts for a maximum of 10 minutes. The team with the most game points at the end wins the match!

1.4 Game Logic

1.4.1 Goal

Score **game points** by conquering enemy boxes and finish the match with more game points than the other team, or win instantly with a Special Maneuver.

1.4.2 ArUco Marker Encoding

Each physical box displays an ArUco marker from the DICT_4X4_50 dictionary. The marker ID encodes the current state of the box:

- **4x** — box is currently **red**.
- **3x** — box is currently **blue**.
- The second digit is the **box ID** (1-6).

Box ID	Color	Marker ID
1	red	41
2	red	42
3	red	43
4	blue	34
5	blue	35
6	blue	36

Table 1: Default ArUco marker IDs per box (match start).

When a box is captured, the displayed marker changes with the box color. Example: box 3 starts red (marker 43), is captured by blue (changes to 33), then recaptured by red (changes back to 43).

1.4.3 Capturing a Box

An enemy box changes to your team's color if:

- your drone hovers over it for at least 2 **seconds**, and
- no defending drone is detected in that box.

A home-zone box changes back to your team's color immediately when recaptured.

After a successful capture, the box is locked for 5 **seconds**.

1 Game Point You get 1 **game point** if:

- your team captures an enemy box,
- the capturing drone returns home, and
- this happens before that box is recaptured by the other team.

If the box is recaptured first, you still get 1 **game point** only if the capturing drone had already reached home; otherwise you get 0 **game points**.

5 Game Points You get 5 **game points** if:

- your team captures an enemy box,
- your team started the game with at least 2 drones,
- at the moment of capture, all drones of your team are **outside** your home zone, and
- then all your drones return home before that box is recaptured.

If one of your drones was still in the home zone at the moment of capture, 5 **game points** are not possible for that attempt. If the box is recaptured before all drones return, 5 **game points** are no longer possible, but you may still get 1 **game point** if the capturing drone had already reached home.

10 Game Points You get 10 **game points** if:

- two different drones of your team capture two different enemy boxes,
- both captures occur within less than 1 **second** of each other, and
- your team started the game with at least 2 drones.

If the timing is too slow, there is no 10-point capture and both captures are treated separately (each can still become 1 or 5 game points). If three boxes are captured almost simultaneously, one valid pair may yield 10 **game points** while the third capture is treated separately.

Home Zone Recapture If your team recaptures a box in your own home zone:

- the box reverts to your team's color immediately, and
- you receive 0 **game points**.

This rule prevents defensive game point farming.

Special Maneuver If one team controls all 6 boxes continuously for at least 5 **seconds**, that team wins the match immediately.

After the instant win is triggered, a final 10-second resolution window remains during which already-open scoring attempts may still finish, but the winning team does not change.

1.4.4 Additional Rules

- One drone can only run one scoring attempt at a time.
- A drone may only start a new attempt after it has been detected back in its own home zone.
- One capture attempt can only score once: 1, 5, or 10 game points. These scores do not stack for the same attempt.
- Game point priority for the same capture event: Special Maneuver > 10 > 5 > 1.

1.5 Tournament

All matches are played head-to-head per the rules in Section 1.3. The tournament is played by **6 teams** in three stages:

- **Opening Round** — All 6 teams compete. The **4 best-ranked** teams advance to the semi-finals.
- **Semi-Finals** — The 4 remaining teams play 2 matches. The **2 winners** advance to the final.
- **Final** — The 2 remaining teams compete for the championship.

Match pairings are determined by the organizers and announced via Discord. In the event that an assigned opponent cannot participate, a replacement team is drawn by lot or, if unavailable, the match is scored as a technical win for the other team.

1.6 Winning the Challenge

The overall winner is the team with the most **challenge points** at the end of the tournament. Challenge points are accumulated from three categories, with a maximum of **25 challenge points** total:

Category	Max. Challenge Points
Match wins (5 challenge points per won match)	15
Jury evaluation (pitch & overall approach)	10
Technical report	5
Total	25

Table 2: Overall challenge scoring overview.

1.6.1 Match Performance

Each won match awards the winning team **5 challenge points**. A team can win at most one match per stage (Opening Round, Semi-Final, Final), for a maximum of **15 challenge points** from match performance.

1.6.2 Jury Evaluation

A jury of experts awards up to **10 challenge points** based on the team’s pitch presentation and overall technical approach demonstrated during the competition.

1.6.3 Technical Report

Each team must submit a technical report that describes their solution. The report is evaluated by the jury for up to **5 challenge points** and must address the following:

- **Localisation** — describe the localisation approach in detail, e.g. Kalman filter state definition, sensor fusion, or other state estimation methods.
- **Flight Control & Trajectory Planning** — explain the flight controller design and trajectory planning strategies used.
- **Swarm Strategy** — describe how the drones coordinate and cooperate as a swarm to achieve the mission objective.

Teams that open-source their code will receive a positive consideration in the technical report evaluation.

The technical report is due **4 days before the final**.

2 Team Regulations

2.1 Team Composition

- Up to 5 team members per team can be present on-site
- New team members are allowed to join but need to be registered with brigkAIR.

2.2 Responsibilities

- Each team must designate at least one Spotter, who assists the pilot in maintaining visual awareness of the drones and ensures safe operation within the arena.
- Each team must also appoint a Safety Officer, who serves as the primary contact for all safety-related matters between the team and the event committee.
 - The Safety Officer must be present during all matches.
 - They hold the authority to initiate an immediate stop of the match ("Killswitch") if any unsafe situation occurs.
 - The Safety Officer is responsible for ensuring that all team members follow the safety protocols at all times.
 - Prior to the Fly-Off, each team must submit information about their Safety Officer (name, role, and contact details) to the event organizers.
- Teams are responsible for virtually validating their solution before the competition to ensure functional and safe system behavior.

2.3 Intellectual Property

- The IP stays with the teams. All intellectual property rights of the hardware and software developed for this challenge remain with the respective teams.
- Exceptions: Any hardware, components, or materials provided by brigkAir remain the property of brigkAir and must be returned after the event if requested.
- Teams are encouraged to open-source their solutions. Doing so may have a positive influence on the jury's evaluation, as it promotes collaboration and knowledge sharing.

2.4 Reporting Requirements

- Each team must submit a technical report of up to five (5) A4 pages, describing the design, implementation, and functionality of their solution.
- In addition, teams are required to prepare a presentation explaining their approach and key technical aspects to the judges during the evaluation phase.
- The presentation shall be five (5) minutes long, prepared in PowerPoint (.ppt) format, and held during both the Qualification and the Final rounds.

2.5 Reimbursement

- Teams are allowed to order parts and materials over time as needed, but must plan carefully since only five reimbursements in total will be processed.
- The combined value of all orders must not exceed €1000.
- Until the final, the hardware remains the property of brigkAIR and is provided to the teams as a permanent loan.

3 Safety Regulations

3.1 Pre-Match and Operational Safety

- The teams are not allowed to start any motors before the official start signal is given by the judges.
- Before the match begins, each team must call "Ready" to the judges to confirm that all drones and operators are in a safe and prepared state.
- Entering the flight arena during the match is strictly prohibited for safety reasons.
- Any violation of these safety rules may result in disqualification from the current match or other disciplinary action as determined by the judges.

- Safety takes priority over scoring — teams must always operate their drones in a controlled and responsible manner.

3.2 Collision Prevention

- Each drone is recommended to be equipped with a spherical protection case, which helps to protect both the drone itself and other drones from damage during the match.
- The protective spheres must be provided by the teams themselves. brigkAir does not supply drone protection equipment.
- If a drone becomes damaged or inoperative during a match, no replacement drone may be added to the team's lineup. The team must continue the match with the remaining functional drones.
- Each team may register one (1) spare drone as a backup before the tournament begins. This spare drone can only be used to replace a malfunctioning unit between matches, not during an active match.
- Teams must always avoid intentional collisions or actions that could cause damage to other drones or the arena. Fair and safe flying behavior is mandatory.
- If a team's number of operational drones drops below two, the team will be disqualified from the current match for safety and fairness reasons.

3.3 Continuity of C2 Link

- A continuous Command and Control (C2) link between the operator and each drone must be maintained at all times.
- If the C2 link is interrupted or lost, an automatic flight termination or controlled landing must be initiated no later than 2 seconds after the loss of connection.
- In the event of a C2 link failure, the match will be paused or safely terminated by the judges. The affected team will not be penalized, as maintaining safety is the highest priority in this competition.
- Once the drone and the area are confirmed safe, the judges will decide whether the match can be resumed or restarted, depending on the situation and remaining time.

3.4 Emergency Shutoff and Kill Switch

- Each drone must be equipped with a physical or virtual kill switch, accessible to the operator or the team's Safety Officer, and its correct functionality must be demonstrated prior to the match.
- Activation of the kill switch must result in an immediate landing or motor shutdown of the drone.
- Each team must have one master button or command that can simultaneously land or shut down all drones of the team in case of an emergency.
- Improper or unsafe operation after a kill-switch event may lead to disqualification, but safe and responsible use of the emergency system will not result in penalties.

4 Technical Regulations

4.1 Drone Requirements

- The Maximum Take-Off Mass (MTOM) of each drone must be below 10 kg. The declared TOM includes all payload, batteries, and protective equipment.
- Each drone swarm may consist of a maximum of 5 drones and at least 2 drones.
- Each drone must be equipped with an RFID marker for automatic identification by the target objects.
- The RFID markers will be provided by brigkAIR at the event and must be properly attached according to the organizer's specifications.

- Teams are responsible for ensuring that the RFID marker is securely mounted and fully functional throughout the competition.

4.2 Swarming Functionality

For the purpose of this challenge, swarming functionality is defined as the coordinated and cooperative behavior of two or more drones that dynamically share information to achieve a common objective.

- Each team must implement and demonstrate a swarming functionality in accordance with the competition’s definition of swarm behavior.
- The teams shall describe and explain their swarming approach in both the technical report and the presentation during the Qualification and Final stages.
- The demonstration must clearly show coordination and communication between multiple drones — i.e., the drones must act as a collective system rather than as independent units.
- The swarming functionality may be realized using different communication or sensing methods, for example:
 - Light-based signaling (e.g., LEDs, optical markers)
 - RF-based communication (e.g., Wi-Fi, LoRa, Bluetooth, UWB)
 - EO-based coordination (electro-optical vision systems, camera tracking)
 - Acoustic or infrared systems, or other innovative interaction methods

4.3 Ground Control Station

- Computing and control do not have to be performed exclusively onboard the drones. A combination of onboard processing and ground-based computation via a Ground Control Station is permitted.
- The swarm behavior may be governed by:
 - A single centralized agent that maintains awareness of the global swarm state, or
 - Distributed agents, where each drone operates semi-independently but coordinates with others through local communication or sensing.
- Teams must describe their chosen swarm architecture and communication topology in their technical documentation.

4.4 Positioning System

The challenge will take place at the **ILA Berlin on Thursday, 11 June 2026**. To improve state estimation in the GPS-denied indoor environment, a camera-based positioning system using ArUco markers can be used.

Two ArUco markers are attached to each pillar in the hangar. Figure 3 illustrates the IDs of the markers that are attached to the pillars, where the lower ArUco marker is mounted at a height of 2m and the upper marker at 4m. In the notation “X/Y”, the first number (X) corresponds to the upper marker at 4m height and the second number (Y) to the lower marker at 2m height.

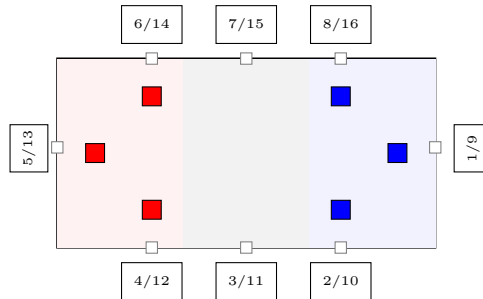


Figure 3: ArUco Marker ID’s

The ArUco markers are from the *DICT*_4 × 4_50 dictionary and have a size of 0.5m x 0.5m. This information enables positioning in the arena with a view of a single marker.

4.5 Arbiter / Judge

- A Ground Truth and Scoring Display visible to both teams is desired. This display should show the current match scores.
- However, no technical solution is provided by the organizers that collects or distributes real-time information on positioning, scoring, or any other referee-related data.
- Teams are therefore encouraged to develop their own sensing and evaluation systems if they wish to monitor swarm position or performance metrics during the match.